



Semester I Examinations 2017/2018

Exam Codes	3BA1, 3BS9, 3MR3, 3BPT1, 4BA4, 4BME1, 4BS2, 1CNS1 1OA1, 1EM1
Exams	Third Arts, Third Science, Third Marine Science, Fourth Arts, Fourth Mathematics and Education, Fourth Science, Master of Science
Module	Applied Statistics I
Module Codes	ST311
External Examiner	Professor S. Wilson
Internal Examiner(s)	Dr. E. Holian Professor J. Hinde
<u>Instructions:</u>	Answer any 3 questions from Questions 1-4 All Questions carry equal marks.
Duration	2 Hours
No. of Pages	21 pages, including this one
School	School of Mathematics, Statistics and Applied Mathematics
Release to Library:	Yes ☒ No ☐
Release in Exam Venue	Yes ☒ No ☐
<u>Requirements:</u>	
Statistical Tables/ Log Tables	Relevant distribution tables are attached to this paper. The <i>New Formulae & Tables</i> are optional.
Other Materials Allowed	A calculator is allowed (non-programmable and not capable of storing text).

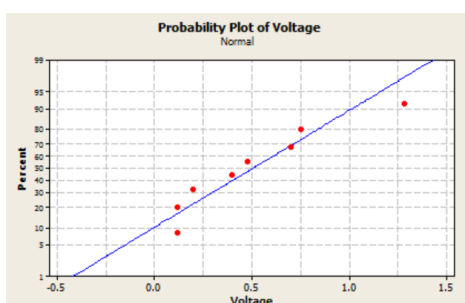
Question 1

Complete parts (a), (b) and (c).

- (a) In the oil industry water that mixes with crude oil in production must be removed. Researchers have observed the voltage required to separate the water from the oil in a sample of 8 trials conducted under controlled conditions. The recorded voltages observed for the sample are: 0.12, 0.12, 0.2, 0.4, 0.48, 0.7, 0.75, 1.28
A summary of descriptive statistics and plots for this sample are provided below:

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Voltage	8	0	0.506	0.140	0.396	0.120	0.140	0.440	0.738	1.280

N for				
Variable	Mode	Mode	Skewness	Kurtosis
Voltage	0.12	2	1.04	0.89



- By comparing the values of the mean, the median and the mode(s) for this sample what does this tell us about the shape of the distribution of the sample? [2 marks]
 - What does the normal probability plot suggest? [2 marks]
 - Construct the letter value display for this sample, including midvalues and spreads. [4 marks]
 - What do the letter values indicate about the sample distribution? [4 marks]
- (b) The researchers wish to determine whether this sample gives evidence that the true typical voltage required under these conditions is greater than 0.2.
- It is suggested that a 1-sample t-test be carried out. The output is included below. Assuming the assumptions of the test hold what can you conclude as a result of this analysis? Reference the output to justify your answer. [3 marks]

One-Sample T: Voltage							
Test of $\mu = 0.2$ vs > 0.2							
Variable	N	Mean	StDev	SE Mean	95% Lower Bound	T	P
Voltage	8	0.506	0.396	0.140	0.241	2.19	0.033

Question 1 continues on the next page

Question 1 continued...

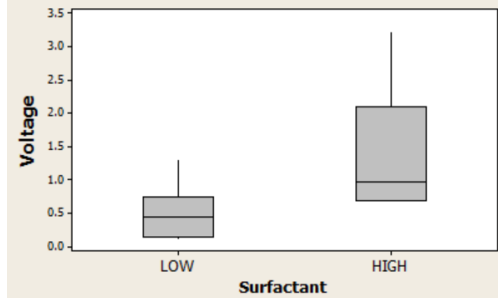
- (ii) Do you have any concerns about applying the 1-sample t-test in this application? If so, explain reasons for your concern. [2 marks]
- (iii) It is suggested that a 1-sample Sign test be carried out. Use the sample to test where there is evidence to suggest $\eta > 0.2$, testing at $\alpha = 0.05$. Include a statement of the hypotheses being tested, an explanation of how the p-value is calculated defining any symbols you use, your decision and conclusion in this application. [7 marks]
- (c) The sample of voltage measurements observed previously were in controlled conditions, in particular, when the surfactant concentration was controlled at a low value of 2%. The researchers wish to compare typical voltage required when the conditions are changed, when surfactant concentration is controlled at a higher value of 4%. A sample of 6 trials are again carried out controlled at this high concentration, the observed voltages in the sample at low surfactant concentration and at high surfactant concentration are:

Low: 0.12, 0.12, 0.2, 0.4, 0.48, 0.7, 0.75, 1.28.

High: 0.68, 0.7, 0.88, 1.04, 1.72, 3.2.

Some descriptive statistics and plots for these samples are also provided.

Variable	N	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
LOW	8	0.506	0.140	0.396	0.120	0.140	0.440	0.738	1.280
HIGH	6	1.370	0.398	0.974	0.680	0.695	0.960	2.090	3.200



The researcher wishes to carry out a hypothesis test to infer whether the samples give evidence that the typical value of voltages for the population of Low surfactant is less than the typical value of voltages for the population of High surfactant.

- (i) It is suggested to you to carry out a Wilcoxon Rank Sum test. Explain why, in this application, this test is more suitable than the 2-sample Z test, or the 2-sample t test. [4 marks]
- (ii) Carry out the test at significance level $\alpha=0.05$. Include a statement of the hypotheses being tested, formulation of the test statistic, the rejection region for the test statistic, and indicate clearly your decision and conclusion from the result of the test. [7 marks]

[Total 35 marks]

Question 2

Complete parts (a), (b) and (c).

- (a) (i) "The true correlation coefficient between two variables X and Y for a population of individuals is known to have value $\rho = 0$. This implies that X and Y are unrelated."
- Is this statement true or false? If you deem the statement to be false give a reason for your answer. [2 marks]
- (ii) A sample of 20 individuals gave a sample correlation coefficient of $r = 0.56$. Carry out a hypothesis test of whether there is evidence of a significant increasing relationship between X and Y in the population. Include clear statements of the hypotheses, the rejection region and conclusion. [6 marks]

(b) Consider a simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$.

- (i) What does the error term, ϵ_i , represent? [2 marks]
- (ii) When carrying out an Analysis of Variance, explain in general what the terms SS_{Total} , SS_{Reg} and SS_{Error} represent. [5 marks]

(c) The following model is being considered in order to model the variability in a response variable Y , using five candidate predictor variables, X_1, X_2, X_3, X_4, X_5 ,

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \epsilon_i.$$

The model is fitted to observations obtained for a sample of 20 individuals, and the results of the fit are provided in the partial printout on the next page.

Use the output to answer the following questions.

- (i) Calculate the coefficient of determination R^2 for this regression fit. Interpret the meaning of this value. [2 marks]
- (ii) Find the missing values in the ANOVA table, and complete the test, at the $\alpha = 0.05$ level, including statements of the hypotheses being tested, a rejection region, a decision and conclusion. (*You do not have to calculate the p-value.*) [6 marks]
- (iii) Complete the missing t-test in the output, i.e. test whether X_2 is a good predictor of Y at the $\alpha = 0.05$ level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion. [5 marks]
- (iv) Use the partial F-test to test the hypothesis : $H_0 : \beta_4 = \beta_5 = 0$ at $\alpha = 0.05$. Interpret your findings. [5 marks]
- (v) The output provides a prediction interval and a confidence interval for the response when $X_1 = 12$, $X_2 = 25$, $X_3 = 5$, $X_4 = 55$, $X_5 = 1$. Explain the difference as to when a prediction interval and a confidence interval applies. [2 marks]

Question 2 continues on the next page

Question 2 continued...

The regression equation is

$$Y = 9.72 + 2.95 X_1 + 2.82 X_2 - 1.20 X_3 + 0.10 X_4 - 11.65 X_5$$

Predictor	Coef	SE Coef	T	P
Constant	9.72	36.940	0.26	0.796
X1	2.95	0.939	3.14	0.007
X2	2.82	1.133	-	-
X3	-1.20	0.468	-2.56	0.022
X4	0.10	0.179	0.56	0.597
X5	-11.65	10.890	-1.07	0.303

S = 20.936 R-Sq = - % R-Sq(adj) = - %

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	-	12810	-	-	-
Residual Error	-	-	-	-	-
Total	-	18947			

Source	DF	Seq SS
X1	1	5906
X2	1	3521
X3	1	2845
X4	1	36
X5	1	502

Predicted Values for New Observations

NewObs	Fit	SE Fit	95% CI	95% PI
1	103.30	7.09	(88.09, 118.51)	(55.89, 150.71)

Values of Predictors for New Observations

NewObs	X1	X2	X3	X4	X5
1	12	25	5	55	1

[Total 35 marks]

Question 3

Complete parts (a) and (b).

- (a) In order to build a model that will estimate the liveweight (LW) of mature cows of a certain breed in a small village in Western Kenya, using measurements of the animal, a sample of 30 animals were selected and observed. For each animal the actual liveweight was recorded, in kgs, and two measurements:

- HG, heartgirth, the girth of the animal at the point of the heart measured with a tape measure, recorded in cms.
- BL, the body length of the animal, measured with a tape measure, recorded in cms.

- (i) The output below is produced by fitting a model where both explanatory variables are used to predict liveweight, $LW_i = \beta_0 + \beta_1 HG_i + \beta_2 BL_i + \epsilon_i$. Discuss the predictive power of the two variables, HG and BL , by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference.

The regression equation is					
LW = - 392 + 1.91 HG + 3.38 BL					
Predictor	Coef	SE Coef	T	P	
Constant	-391.60	71.20	-5.50	0.000	
HG	1.909	2.293	0.83	0.412	
BL	3.381	2.915	1.16	0.256	
S = 18.7810 R-Sq = 82.1% R-Sq(adj) = 80.7%					
Analysis of Variance					
Source	DF	SS	MS	F	P
Regression	2	43566	21783	61.76	0.000
Residual Error	27	9524	353		
Total	29	53090			
Source	DF	Seq SS			
HG	1	43092			
BL	1	475			

[12 marks]

- (ii) Explain the role of R^2 and adjusted R^2 in finding the parsimonious model. Comment on the values provided for three models below to help illustrate your answer.

	R-Sq	R-Sq(adj)
Model 1: $LW_i = \beta_0 + \beta_1 HG_i + \epsilon_i$	81.2	80.5
Model 2: $LW_i = \beta_0 + \beta_1 BL_i + \epsilon_i$	81.6	80.9
Model 3: $LW_i = \beta_0 + \beta_1 HG_i + \beta_2 BL_i + \epsilon_i$	82.1	80.7

[5 marks]

Question 3 continues on the next page

Question 3 continued...

- (iii) The output below is produced by fitting a model where bodylength is used to predict liveweight, $LW_i = \beta_0 + \beta_1 BL_i + \epsilon_i$.

The output references individuals in the sample that have a large standardised residual.

The last observation, with predictor value $BL = 106$ and observed response $LW = 289.15$, has some output values missing.

Find the value of the fitted response $\hat{Y}$ and the fitted residual $\hat{\epsilon}$ for this individual, and thus show how the standardised residual of 2.06 is arrived at.

Why have these observations been flagged as unusual observations?

[5 marks]

The regression equation is

$$LW = -361 + 5.77 BL$$

Predictor	Coef	SE Coef	T	P
Constant	-361.00	60.66	-5.95	0.000
BL	5.77	0.5177	11.14	0.000

S = 18.68 R-Sq = 81.6% R-Sq(adj) = 80.9%

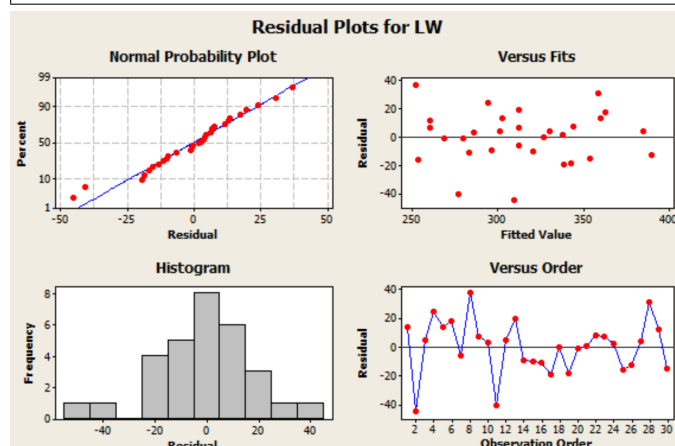
Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	43322	43322	124.18	0.000
Residual Error	28	9768	349		
Total	29	53090			

Unusual Observations

Obs	BL	LW	Fit	SE Fit	Residual	St Resid
2	116	264.90	309.91	3.43	-45.01	-2.45R
8	111	236.14	276.65	4.78	-40.51	-2.24R
11	106	289.15	-	6.53	-	2.06R

R denotes an observation with a large standardized residual.



Question 3 continues on the next page

Question 3 continued...

- (b) A researcher wishes to explore the variability in annual salary amongst those in the field of social work. A random sample of 50 social workers was selected and their current salary and number of years experience recorded.

Output 1 displays the results and diagnostic plots for a simple linear regression fit to the data. The output of the routine to fit the Box-Cox transformation is also included.

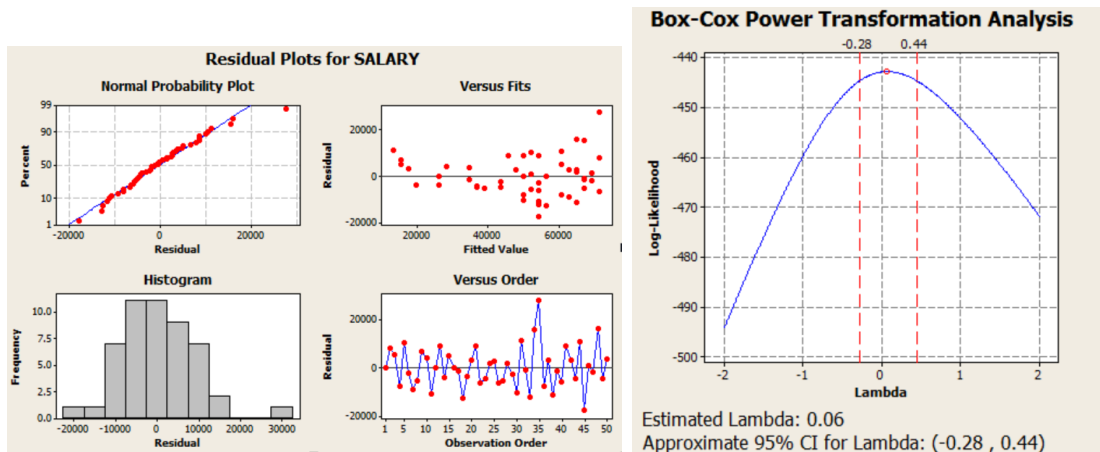
- (i) Comment on the validity of the assumptions for this regression fit. [5 marks]
- (ii) What does the output of the Box-Cox transformation indicate? [4 marks]
- (iii) A simple linear regression using *EXP* as the predictor was fitted to the transformed response variable $\ln(\text{SALARY})$ giving output 2 (see next page). Using this model predict the salary of a social worker who has 10 years experience. [4 marks]

Output 1:

The regression equation is
 $\text{SALARY} = 11369 + 2141 \text{ EXP}$

Predictor	Coef	SE Coef	T	P
Constant	11369	3160	3.60	0.001
EXP	2141	160.8	13.31	0.000

S = 8642.44 R-Sq = 78.7% R-Sq(adj) = 78.2%



Question 3 continues on the next page

Question 3 continued...

Output 2:

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The regression equation is
ln(SALARY) = 9.84 + 0.05 EXP

Predictor    Coef    SE Coef      T      P
Constant     9.84    0.05636   174.63  0.000
EXP           0.05    0.002868   17.43  0.000

S = 0.154113   R-Sq = 86.4%   R-Sq(adj) = 86.1%
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[Total 35 marks]

Question 4

Complete parts (a), (b) and (c).

- (a) A researcher compared the weights of three different fish species.

Indicator variables are defined for the three species:

$$\begin{aligned} IndCATF : & \begin{cases} 1 & \text{if the fish observed is species 1, a catfish} \\ 0 & \text{otherwise} \end{cases} \\ IndBASS : & \begin{cases} 1 & \text{if the fish observed is species 2, a bass} \\ 0 & \text{otherwise} \end{cases} \\ IndBUFF : & \begin{cases} 1 & \text{if the fish observed is species 3, a buffalo} \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

and the model, $Weight_i = \beta_0 + \beta_1 IndBASS_i + \beta_2 IndBUFF_i + \epsilon_i$, is fitted to the data, giving:

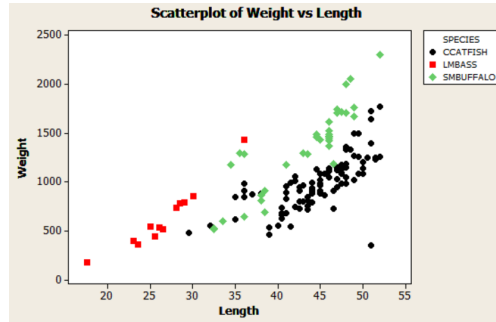
The regression equation is				
Weight = 989 - 360 IndBASS + 368 IndBUFF				
Predictor	Coef	SE Coef	T	P
Constant	988.60	32.88	30.07	0.000
IndBASS	-359.60	98.17	-3.66	0.000
IndBUFF	367.82	62.71	5.86	0.000
S = 320.440 R-Sq = 29.0% R-Sq(adj) = 28.0%				

- i) Use the fitted model to predict weight for:
- a fish of species BASS.
 - a fish of species BUFF.
 - a fish of species CATF.
- [3 marks]
- ii) Interpret the meaning of the estimated coefficients of $IndBASS$ and $IndBUFF$, and comment on their significance. Ensure you state the hypotheses being tested when referring to the results of the output.
- [6 marks]
- iii) Write down an alternative model proposal that you would recommend be fitted to the observed data that would help assess the comparison of weight between each of the species, CATF and BASS, compared with the BUFF species.
- [2 marks]

Question 4 continues on the next page

Question 4 continued...

- (b) In an attempt to control for substantial variability in weight, the length of each fish was measured as a covariate. A plot of the observed relationship of weight with length is provided, with an indication to the different species.



The regression analysis below models the response, *Weight*, by the predictor *Length*, and by the indicator variables *IndCATF*, *IndBASS*, as defined above.

The regression equation is

$$\text{Weight} = -889 + 52.1 \text{ Length} - 451 \text{ IndCATF} + 136 \text{ IndBASS}$$

Predictor	Coef	SE Coef	T	P
Constant	-889.3	155.7	-5.71	0.000
Length	52.075	3.525	14.77	0.000
IndCATF	-450.66	39.66	-11.36	0.000
IndBASS	136.16	88.82	1.53	0.128

S = 200.609 R-Sq = 72.4% R-Sq(adj) = 71.8%

- Interpret the meaning of the fitted coefficients of *Length*, *IndCATF* and *IndBASS*, in this model. [5 marks]
 - For this model, write down the estimated relationship between weight and length for each of the three species, CATF, BASS and BUFF. [3 marks]
 - What assumption is being made about the relationship of weight and length for the three different species in this model? [2 marks]
- (c) Two interaction terms were added to the model:
Length.IndCATF, calculated as $(\text{Length}_i) \times (\text{IndCATF}_i)$ for each individual,
Length.IndBASS, calculated as $(\text{Length}_i) \times (\text{IndBASS}_i)$ for each individual,
giving the following output:

Question 4 continues on the next page

Question 4 continued...

The regression equation is
 Weight = - 1717 + 71.3 Length + 907 IndCATF + 494 IndBASS
 - 31.0 Length.IndCATF - 1.5 Length.IndBASS

Predictor	Coef	SE Coef	T	P
Constant	-1717.1	255.7	-6.71	0.000
Length	71.270	5.885	12.11	0.000
IndCATF	906.9	318.5	2.85	0.005
IndBASS	493.9	426.3	1.16	0.249
Length.IndCATF	-31.042	7.244	-4.29	0.000
Length.IndBASS	-1.49	13.99	-0.11	0.916

S = 188.486 R-Sq = 76.0% R-Sq(adj) = 75.1%

- i) For this model, write down the estimated relationship between weight and length for each of the three species. [4 marks]
- ii) Interpret the meaning of the fitted coefficients of *Length.IndCATF* and *Length.IndBASS* in this model. [5 marks]
- iii) Determine if the terms *Length.IndCATF* and *Length.IndBASS* are significant, using the results of the individual t-tests in this analysis. Ensure you state the hypotheses being tested when referring to the results of the output. [5 marks]

[Total 35 marks]

ST311 Formulae

Some useful z-scores:

$$Z_{0.1} = 1.28$$

$$Z_{0.05} = 1.64$$

$$Z_{0.025} = 1.96$$

$$Z_{0.01} = 2.33$$

$$Z_{0.005} = 2.58$$

Some ranges for the standard normal distribution:

$$1.35 \quad \text{middle } 50\% : Z_{0.75} - Z_{0.25}$$

$$2.30 \quad \text{middle } 75\% : Z_{0.875} - Z_{0.125}$$

$$3.07 \quad \text{middle } 87.5\% : Z_{0.9375} - Z_{0.0625}$$

$$3.73 \quad \text{middle } 93.75\% : Z_{0.96875} - Z_{0.03125}$$

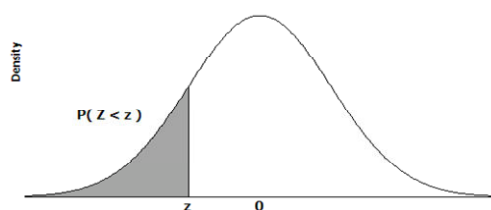
$$4.31 \quad \text{middle } 96.875\% : Z_{0.984375} - Z_{0.015625}$$

$$4.84 \quad \text{middle } 98.4375\% : Z_{0.9921875} - Z_{0.0078125}$$

Some formulae that may be useful when carrying out inference with parametric procedures:

- Inference for population correlation coefficient ρ : $TS = r\sqrt{\frac{n-2}{1-r^2}}$
- Partial F-test : $TS = \frac{(\text{additional SSreg})/q}{MS_{error}}$

Table: Cumulative Probabilities for the Standard Normal Distribution, $Z \sim \text{Normal}(0,1)$.



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table: Cumulative Probabilities for the Standard Normal Distribution, $Z \sim \text{Normal}(0,1)$.

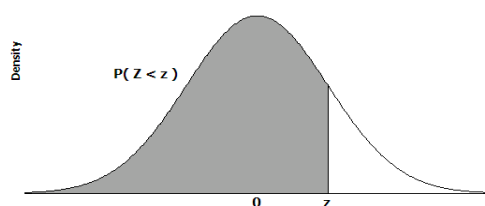
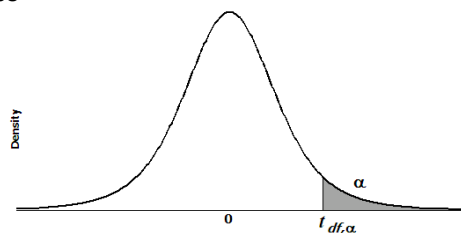
[illegible]

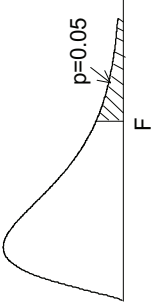
Table: t distribution critical values



df	Upper tail probability											
	0.25	0.20	0.15	0.10	0.05	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
Z*	0.674	0.842	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.291

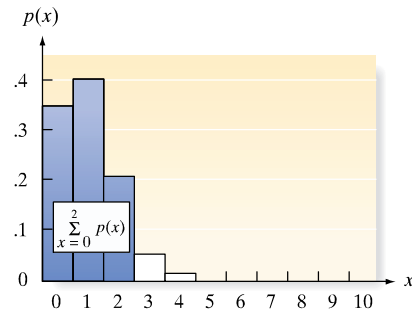
F distribution for $p = 0.05$

For various pairs of degrees of freedom (v_1, v_2), the tabled entries represent the critical values of F above which the proportion $p = 0.05$ of the distribution falls. (Example, for $df = 3, 8$ and $F = 2.92$ is exceeded by $p = 0.05$ of the population.



$p = 0.05$		Degrees of freedom for the numerator v_1																		
v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞		
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317		
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496		
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526		
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628		
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365		
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669		
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230		
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928		
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707		
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538		
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404		
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296		
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206		
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131		
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066		
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010		
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960		
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917		
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878		
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843		
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812		
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783		
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757		
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733		
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622		
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509		
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.399		
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254		
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000		

TABLE II Binomial Probabilities



Tabulated values are $\sum_{x=0}^k p(x)$. (Computations are rounded at the third decimal place.)

a. $n = 5$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.951	.774	.590	.328	.168	.078	.031	.010	.002	.000	.000	.000	.000
1	.999	.977	.919	.737	.528	.337	.188	.087	.031	.007	.000	.000	.000
2	1.000	.999	.991	.942	.837	.683	.500	.317	.163	.058	.009	.001	.000
3	1.000	1.000	1.000	.993	.969	.913	.812	.663	.472	.263	.081	.023	.001
4	1.000	1.000	1.000	1.000	.998	.990	.969	.922	.832	.672	.410	.226	.049

b. $n = 6$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.941	.735	.531	.262	.118	.047	.016	.004	.001	.000	.000	.000	.000
1	.999	.967	.886	.655	.420	.233	.109	.041	.011	.002	.000	.000	.000
2	1.000	.998	.984	.901	.744	.544	.344	.179	.070	.017	.001	.000	.000
3	1.000	1.000	.999	.983	.930	.821	.656	.456	.256	.099	.016	.002	.000
4	1.000	1.000	1.000	.998	.989	.959	.891	.767	.580	.345	.114	.033	.001
5	1.000	1.000	1.000	1.000	.999	.996	.984	.953	.882	.738	.469	.265	.059

c. $n = 7$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.932	.698	.478	.210	.082	.028	.008	.002	.000	.000	.000	.000	.000
1	.998	.956	.850	.577	.329	.159	.063	.019	.004	.000	.000	.000	.000
2	1.000	.996	.974	.852	.647	.420	.227	.096	.029	.005	.000	.000	.000
3	1.000	1.000	.997	.967	.874	.710	.500	.290	.126	.033	.003	.000	.000
4	1.000	1.000	1.000	.995	.971	.904	.773	.580	.353	.148	.026	.004	.000
5	1.000	1.000	1.000	1.000	.996	.981	.937	.841	.671	.423	.150	.044	.002
6	1.000	1.000	1.000	1.000	1.000	.998	.992	.972	.918	.790	.522	.302	.068

(continued)

TABLE II (continued)

d. $n = 8$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.923	.663	.430	.168	.058	.017	.004	.001	.000	.000	.000	.000	.000
1	.997	.943	.813	.503	.255	.106	.035	.009	.001	.000	.000	.000	.000
2	1.000	.994	.962	.797	.552	.315	.145	.050	.011	.001	.000	.000	.000
3	1.000	1.000	.995	.944	.806	.594	.363	.174	.058	.010	.000	.000	.000
4	1.000	1.000	1.000	.990	.942	.826	.637	.406	.194	.056	.005	.000	.000
5	1.000	1.000	1.000	.999	.989	.950	.855	.685	.448	.203	.038	.006	.000
6	1.000	1.000	1.000	1.000	.999	.991	.965	.894	.745	.497	.187	.057	.003
7	1.000	1.000	1.000	1.000	1.000	.999	.996	.983	.942	.832	.570	.337	.077

e. $n = 9$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.914	.630	.387	.134	.040	.010	.002	.000	.000	.000	.000	.000	.000
1	.997	.929	.775	.436	.196	.071	.020	.004	.000	.000	.000	.000	.000
2	1.000	.992	.947	.738	.463	.232	.090	.025	.004	.000	.000	.000	.000
3	1.000	.999	.992	.914	.730	.483	.254	.099	.025	.003	.000	.000	.000
4	1.000	1.000	.999	.980	.901	.733	.500	.267	.099	.020	.001	.000	.000
5	1.000	1.000	1.000	.997	.975	.901	.746	.517	.270	.086	.008	.001	.000
6	1.000	1.000	1.000	1.000	.996	.975	.910	.768	.537	.262	.053	.008	.000
7	1.000	1.000	1.000	1.000	1.000	.996	.980	.929	.804	.564	.225	.071	.003
8	1.000	1.000	1.000	1.000	1.000	1.000	.998	.990	.960	.866	.613	.370	.086

f. $n = 10$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.904	.599	.349	.107	.028	.006	.001	.000	.000	.000	.000	.000	.000
1	.996	.914	.736	.376	.149	.046	.011	.002	.000	.000	.000	.000	.000
2	1.000	.988	.930	.678	.383	.167	.055	.012	.002	.000	.000	.000	.000
3	1.000	.999	.987	.879	.650	.382	.172	.055	.011	.001	.000	.000	.000
4	1.000	1.000	.998	.967	.850	.633	.377	.166	.047	.006	.000	.000	.000
5	1.000	1.000	1.000	.994	.953	.834	.623	.367	.150	.033	.002	.000	.000
6	1.000	1.000	1.000	.999	.989	.945	.828	.618	.350	.121	.013	.001	.000
7	1.000	1.000	1.000	1.000	.998	.988	.945	.833	.617	.322	.070	.012	.000
8	1.000	1.000	1.000	1.000	1.000	.998	.989	.954	.851	.624	.264	.086	.004
9	1.000	1.000	1.000	1.000	1.000	1.000	.999	.994	.972	.893	.651	.401	.096

(continued)

TABLE II (continued)

g. $n = 15$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.860	.463	.206	.035	.005	.000	.000	.000	.000	.000	.000	.000	.000
1	.990	.829	.549	.167	.035	.005	.000	.000	.000	.000	.000	.000	.000
2	1.000	.964	.816	.398	.127	.027	.004	.000	.000	.000	.000	.000	.000
3	1.000	.995	.944	.648	.297	.091	.018	.002	.000	.000	.000	.000	.000
4	1.000	.999	.987	.838	.515	.217	.059	.009	.001	.000	.000	.000	.000
5	1.000	1.000	.998	.939	.722	.403	.151	.034	.004	.000	.000	.000	.000
6	1.000	1.000	1.000	.982	.869	.610	.304	.095	.015	.001	.000	.000	.000
7	1.000	1.000	1.000	.996	.950	.787	.500	.213	.050	.004	.000	.000	.000
8	1.000	1.000	1.000	.999	.985	.905	.696	.390	.131	.018	.000	.000	.000
9	1.000	1.000	1.000	1.000	.996	.966	.849	.597	.278	.061	.002	.000	.000
10	1.000	1.000	1.000	1.000	.999	.991	.941	.783	.485	.164	.013	.001	.000
11	1.000	1.000	1.000	1.000	1.000	.998	.982	.909	.703	.352	.056	.005	.000
12	1.000	1.000	1.000	1.000	1.000	1.000	.996	.973	.873	.602	.184	.036	.000
13	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.833	.451	.171	.010
14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.794	.537	.140

h. $n = 20$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.818	.358	.122	.012	.001	.000	.000	.000	.000	.000	.000	.000	.000
1	.983	.736	.392	.069	.008	.001	.000	.000	.000	.000	.000	.000	.000
2	.999	.925	.677	.206	.035	.004	.000	.000	.000	.000	.000	.000	.000
3	1.000	.984	.867	.411	.107	.016	.001	.000	.000	.000	.000	.000	.000
4	1.000	.997	.957	.630	.238	.051	.006	.000	.000	.000	.000	.000	.000
5	1.000	1.000	.989	.804	.416	.126	.021	.002	.000	.000	.000	.000	.000
6	1.000	1.000	.998	.913	.608	.250	.058	.006	.000	.000	.000	.000	.000
7	1.000	1.000	1.000	.968	.772	.416	.132	.021	.001	.000	.000	.000	.000
8	1.000	1.000	1.000	.990	.887	.596	.252	.057	.005	.000	.000	.000	.000
9	1.000	1.000	1.000	.997	.952	.755	.412	.128	.017	.001	.000	.000	.000
10	1.000	1.000	1.000	.999	.983	.872	.588	.245	.048	.003	.000	.000	.000
11	1.000	1.000	1.000	1.000	.995	.943	.748	.404	.113	.010	.000	.000	.000
12	1.000	1.000	1.000	1.000	.999	.979	.868	.584	.228	.032	.000	.000	.000
13	1.000	1.000	1.000	1.000	1.000	.994	.942	.750	.392	.087	.002	.000	.000
14	1.000	1.000	1.000	1.000	1.000	.998	.979	.874	.584	.196	.011	.000	.000
15	1.000	1.000	1.000	1.000	1.000	1.000	.994	.949	.762	.370	.043	.003	.000
16	1.000	1.000	1.000	1.000	1.000	1.000	.999	.984	.893	.589	.133	.016	.000
17	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.996	.965	.794	.323	.075	.001
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.992	.931	.608	.264	.017
19	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.988	.878	.642	.182

(continued)

TABLE XIV Critical Values of T_L and T_U for the Wilcoxon Rank Sum Test: Independent Samples

Test statistic is the rank sum associated with the smaller sample (if equal sample sizes, either rank sum can be used).

a. $\alpha = .025$ one-tailed; $\alpha = .05$ two-tailed

$n_2 \backslash n_1$	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	5	16	6	18	6	21	7	23	7	26	8	28	8	31	9	33
4	6	18	11	25	12	28	12	32	13	35	14	38	15	41	16	44
5	6	21	12	28	18	37	19	41	20	45	21	49	22	53	24	56
6	7	23	12	32	19	41	26	52	28	56	29	61	31	65	32	70
7	7	26	13	35	20	45	28	56	37	68	39	73	41	78	43	83
8	8	28	14	38	21	49	29	61	39	73	49	87	51	93	54	98
9	8	31	15	41	22	53	31	65	41	78	51	93	63	108	66	114
10	9	33	16	44	24	56	32	70	43	83	54	98	66	114	79	131

b. $\alpha = .05$ one-tailed; $\alpha = .10$ two-tailed

$n_2 \backslash n_1$	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	6	15	7	17	7	20	8	22	9	24	9	27	10	29	11	31
4	7	17	12	24	13	27	14	30	15	33	16	36	17	39	18	42
5	7	20	13	27	19	36	20	40	22	43	24	46	25	50	26	54
6	8	22	14	30	20	40	28	50	30	54	32	58	33	63	35	67
7	9	24	15	33	22	43	30	54	39	66	41	71	43	76	46	80
8	9	27	16	36	24	46	32	58	41	71	52	84	54	90	57	95
9	10	29	17	39	25	50	33	63	43	76	54	90	66	105	69	111
10	11	31	18	42	26	54	35	67	46	80	57	95	69	111	83	127

Source: From F. Wilcoxon, and R. A. Wilcox. "Some Rapid Approximate Statistical Procedures," 1964, 20–23. Courtesy of Lederle Laboratories Division of American Cyanamid Company, Madison, NJ.